

# KeepKey Firmware Test Report

**Firmware 7.19.0 | 2026-09-17 20:40 | 20/20 PASSED**

Scope: this report is a curated catalog of 20 tests. The CI run collected 25 (0 of them native firmware unit tests); 0 SKIPPED and did not execute, usually because the emulator predates the firmware the test targets -- a skip is not evidence the feature works. Absence from this report is NOT evidence that a feature is untested -- check the JUnit artifacts.

## Sections

[ER ERC-7730 v2 Certified Clear-Signing Conformance \[NEW\]](#) -- 20/20 passed

## ER. ERC-7730 v2 Certified Clear-Signing Conformance [NEW]

ERC-7730 definitions are compiled into a bounded canonical program, authenticated by KeepKey's certified delegation hierarchy, and interpreted against the exact calldata or EIP-712 document being signed. The host supplies types, labels and formatting policy; it never supplies authoritative decoded transaction values. A certified-definition failure refuses signing and cannot silently downgrade to a less-specific certified display.

### User Flow

AUTHENTICATION AND REPLAY BOUNDARY:

- K773 envelope: version + purpose + canonical C773 program + sorted Merkle proof + 139-byte KeepKey delegation certificate + compact secp256k1 signature.
- Purpose digest: SHA256("KEEPKEY:ERC7730:CATALOG\0" || catalog\_root).
- Definition id commits to the complete envelope; offset-zero replay is re-hashed before staged interpreter output becomes authoritative.
- Catalog lookup binds kind, chain, target and selector/typeHash. Unknown, ambiguous, reordered, oversized and recursively repeated definitions fail closed.

CANONICAL PROGRAM AND DEVICE INTERPRETER:

- 179-byte header; sorted UTF-8 strings; canonical recursive ABI/EIP-712 node tree; typed structured/container/literal paths; typed literals and conditions; formatter, display, deployment/domain binding and resource sections.
- Recursive tuples and arrays through depth 12, 64 ABI nodes, 64 aggregate array elements, negative indices, half-open slices, nested array frames and separators.
- Display bytecode: intent, atomic interpolation fallback, fields, nested groups, arrays, embedded calls and explicit end; all forward/back links and depth are recomputed.
- Conditions: always, never, optional, empty/not-empty, in/not-in and must-match.
- Formatters: raw, native/token amounts, NFT, date, duration, unit, enum, chain, address name, ticker, ERC-7930 interoperable address, embedded calldata and encrypted fallback.
- Signed token/network records establish ticker, decimals and native currency. Live results may annotate but cannot replace a device-decoded value.

REAL-WORLD AND EXHAUSTIVE EVIDENCE:

- Exact KeepKey SDK THORChain router deposit calldata, including vault, asset, amount and memo, is selector-bound and firmware-validated.
- Official Uniswap Router02 signed transaction and Permit2 typed-data fixtures bind the expected selector, target, chain, typeHash and domain constraints.
- Every one of 1,450 calldata formats exposed after bounded include expansion in the official ERC-7730 registry compiles and reaches the firmware catalog verifier.
- Deep ParaSwap Augustus multiSwap/megaSwap definitions exercise depths 9 and 11 while the compacted signing workflow remains exactly 4,096 bytes of fixed SRAM.
- The SDK/Vault consumer path accepts the same signed catalog on ethSignTransaction and from Relay/ShapeShift quote payloads, preserves it through swap construction, preloads the primary envelope, and answers device-driven definition requests in 1,024-byte chunks. Missing, malformed, mismatched, oversized and over-depth requests abort.
- Consumer evidence: hdwallet PR #3 passed build/lint and 8 focused catalog/policy tests; keepkey-vault PR #444 passed 59 focused REST schema and real Relay swap parser tests, Linux/macOS release builds and the packaged Intel macOS smoke test before merge.
- Device evidence: 133 focused firmware tests pass: ABI (5), streamed ABI (7), catalog (19), conditions (6), canonical EIP-712 (33), formatting (8), program readers (20), transaction continuation (3), and authenticated workflow execution (32).
- Host evidence: 25 focused python-keepkey tests pass for deterministic compilation, official-registry traversal, signed envelopes, request policy and protobuf transport.

COMPLETE ACTIVE-V2 SUPPORT BREAKDOWN:

- Descriptor/context: schema-major enforcement; bounded includes and deterministic merge; owner/info/constants/maps/enums; canonical selector and EIP-712 encodeType/typeHash; exact chain/deployment/domain/contract/selector binding; unknown selectors refuse.
- ABI/value model: every legal int width; address/bool/bytesN/bytes/string; fixed and dynamic arrays; static/dynamic tuples; depth-12 recursion; negative index, full-array and bounded-slice paths; transaction, structured and literal containers.
- References: #/\$/@ names, fields, groups, constants, maps, enums, tokens and networks are host-resolved to typed canonical ids; the device rechecks type, range and section links.
- Display: string/object intents, atomic interpolated intent fallback, ordered fields, recursive groups, authenticated array iteration and separators.

- Visibility/assertions: always, never, empty, not-empty, in, not-in and must-match; condition operands are captured from device-decoded calldata or streamed typed data.
- Formatting: raw integer/string/bytes/address; native/token amounts and thresholds; NFT; date, duration, unit, enum, chain id, trusted-name fallback, ticker and ERC-7930.
- Nested/special: embedded calldata requests a separately certified definition with a depth-4/cycle boundary; encrypted data never claims plaintext without verified proof; ERC-4337/EIP-5792 shapes are handled as ordinary bounded EIP-712 structs/arrays.
- Failure policy: dirty padding, gaps/overlap/trailing bytes, bad offsets, wrong context, missing definitions, invalid proof/certificate/purpose, resource excess, ambiguity and host/device value conflicts all refuse signing; certified failure never downgrades.

LIMIT OF THIS SECTION: catalog-verifier success ends at the expected UNTRUSTED result when tests use a synthetic certificate. A trusted-success emulator capture additionally requires a certificate issued by the KeepKey root for the test delegate.

## Tests: 20/20 -- ALL PASSED

### ✓ER1 test\_official\_registry\_all\_calldata\_formats\_reach\_firmware

All 1,450 official calldata formats reach firmware (test\_erc7730\_compiler.py)  
 Recursively merges bounded includes, compiles every applicable registry format and feeds each complete canonical program through the firmware catalog verifier.

### ✓ER2 test\_compiles\_and\_checks\_exact\_keepkey\_sdk\_thorchain\_swap

Exact KeepKey SDK THORChain swap fixture (test\_erc7730\_compiler.py)  
 Checks selector, vault, asset, amount, dynamic memo and transaction value before firmware validation.

### ✓ER3 test\_compiles\_official\_uniswap\_tuple\_fixture\_through\_firmware

Official Uniswap signed transaction (test\_erc7730\_compiler.py)  
 Binds the real Router02 target, chain, selector and published transaction hash.

### ✓ER4 test\_compiles\_official\_uniswap\_eip712\_fixture\_through\_firmware

Official Permit2 EIP-712 fixture (test\_erc7730\_compiler.py)  
 Proves canonical encodeType/typeHash, domain constraints and signed token/network metadata reach the firmware parser.

### ✓ER5 test\_loads\_bounded\_includes\_and\_compiles\_array\_backed\_group

References, includes and grouped arrays (test\_erc7730\_compiler.py)  
 Exercises deterministic include merge, shared field references and a group executed inside an authenticated array frame.

### ✓ER6 test\_compiles\_recursive\_array\_iteration\_frames

Recursive ABI arrays and separators (test\_erc7730\_compiler.py)  
 Builds nested array begin/end links and validates their declared resource depth.

### ✓ER7 test\_compiles\_interpolated\_intent\_and\_metadata\_enum

Atomic interpolation and enum maps (test\_erc7730\_compiler.py)  
 Every interpolated operand is also an unconditional field; failure selects the fallback intent atomically.

### ✓ER8 test\_compiles\_typed\_if\_not\_in\_and\_must\_match\_conditions

Typed visibility and must-match conditions (test\_erc7730\_compiler.py)  
 Typed literal sets control visibility; must-match is a signing assertion.

### ✓ER9 test\_signed\_catalog\_envelope\_commits\_program\_proof\_and\_certificate

Certified Merkle catalog envelope (test\_erc7730\_catalog.py)  
 Verifies deterministic root construction, purpose digest, signature and exact wire layout.

### ✓ER10 test\_catalog\_refuses\_unknown\_ambiguous\_and\_malformed\_requests

Catalog lookup fails closed (test\_erc7730\_catalog.py)  
 Unknown selectors, oversized chunks, excessive recursion and ambiguous tuples are rejected.

### ✓ER11 test\_definition\_chunk\_round\_trip\_is\_byte\_exact

Canonical protocol transport is byte exact (test\_erc7730\_protocol\_bindings.py)  
 Protobuf bindings preserve definition id, offsets, total length and envelope bytes.

### ✓ER12 test\_token\_amount\_without\_token\_uses\_firmware\_raw\_fallback

Unknown tokens never become trusted labels (test\_erc7730\_compiler.py)  
 A missing token reference displays the device-decoded raw integer and forbids token-specific auxiliary metadata.

### ✓ER13 test\_parses\_recursive\_tuple\_and\_array\_signature

Canonical recursive ABI signature parsing (test\_erc7730\_compiler.py)  
 Parses nested tuple and array fragments into the exact bounded forward tree consumed by the firmware interpreter.

### ✓ER14 test\_compiles\_deterministic\_canonical\_calldata\_program

Compilation is byte-for-byte deterministic (test\_erc7730\_compiler.py)  
 Equivalent descriptor input produces one canonical program, selector and source-binding identity rather than host-dependent wire bytes.

#### ✓ER15 test\_compiles\_array\_iteration\_separator\_and\_optional\_visibility

Array iteration, separators and optional visibility (test\_erc7730\_compiler.py)  
Compiles array-backed fields with authenticated length, separator control flow and ifEmpty/ifNotEmpty behavior.

#### ✓ER16 test\_compiles\_nested\_field\_group\_with\_balanced\_links

Recursive groups have balanced control flow (test\_erc7730\_compiler.py)  
Every group call/return and display-program forward link is recomputed and bounded.

#### ✓ER17 test\_catalog\_serves\_exact\_lookup\_and\_id\_replays

Exact-context lookup and authenticated replay (test\_erc7730\_catalog.py)  
Serves only the requested identity tuple and re-authenticates offset-zero definition-id replays before interpreter output becomes authoritative.

#### ✓ER18 test\_preload\_streams\_and\_checks\_every\_acknowledgement

Preload transport checks every acknowledgement (test\_erc7730\_catalog.py)  
Streams bounded chunks and refuses reordered, missing or mismatched device acknowledgements.

#### ✓ER19 test\_erc7730\_message\_ids\_and\_mapping

Canonical message ids are stable (test\_erc7730\_protocol\_bindings.py)  
Pins request, chunk and acknowledgement messages to the canonical protocol mapping.

#### ✓ER20 test\_lookup\_kinds\_encode\_expected\_context

Every definition kind carries its security context (test\_erc7730\_protocol\_bindings.py)  
Calldata, EIP-712, token and network lookups encode only their required chain, target, selector or type-hash identity fields.